

Distributed Computing Meets Game Theory: Robust Mechanisms for Rational Secret Sharing and Multiparty Computation

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Abstract—Traditional distributed computing protocols assume the presence of honest and malicious participants. However, many real-world systems involve rational agents who act according to their incentives. This paper studies the intersection of distributed computing and game theory by introducing robust mechanisms for rational secret sharing and secure multiparty computation (MPC). We analyze *k-resilient Nash equilibria*, which tolerate deviations by coalitions of up to k players. Our results demonstrate that strongly resilient equilibria can be achieved for secret sharing and multiparty computation even with rational agents. We further extend the model to tolerate players with unexpected utilities and show that protocols with mediators can be simulated using cheap-talk communication. These results provide a theoretical framework for secure distributed computation in environments with strategic participants.

I. INTRODUCTION

Distributed computing and cryptographic protocols traditionally classify participants as either honest or malicious. Honest participants strictly follow the protocol, while malicious ones attempt to disrupt it. However, this binary model fails to capture many real-world systems where agents act rationally, aiming to maximize their own benefit.

Game theory offers a natural framework to model such behavior. Instead of assuming malicious adversaries, we consider *rational agents* whose actions depend on incentives. Earlier work showed that rational secret sharing protocols can achieve Nash equilibrium, ensuring that no single player benefits from deviating.

However, Nash equilibrium only protects against unilateral deviations. In distributed systems, players may form coalitions and coordinate their actions. To address this limitation, we study *k-resilient equilibria*, where no coalition of size at most k can benefit from deviating from the prescribed protocol.

This paper investigates mechanisms for secret sharing and multiparty computation under rational behavior. Our main contributions include:

- Introducing mechanisms that achieve strong coalition resilience.
- Providing protocols that work even for two rational agents.
- Extending models to tolerate players with arbitrary or unexpected utilities.
- Demonstrating how mediator-based equilibria can be simulated using cheap talk.

These results provide stronger guarantees for distributed computation in strategic environments.

II. GAME-THEORETIC MODEL

We consider a system of n players $N = \{1, 2, \dots, n\}$ participating in a distributed protocol. Each player selects a strategy σ_i that determines its behavior. The vector $\sigma = (\sigma_1, \dots, \sigma_n)$ represents the joint strategy.

The utility of player i under strategy profile σ is denoted by $u_i(\sigma)$.

A. Nash Equilibrium

A joint strategy σ is a Nash equilibrium if no player can improve its utility by unilaterally changing its strategy:

$$u_i(\sigma) \geq u_i(\sigma'_i, \sigma_{-i})$$

(1) for all players i and alternative strategies σ'_i .

While NashF equilibrium ensures stability against individual deviations, it does not address coordinated deviations by coalitions.

B. k-Resilient Equilibrium

To capture coalition behavior, we define *k-resilient equilibrium*. A strategy profile σ is *k-resilient* if no coalition C of size $|C| \leq k$ can jointly deviate and improve the utility of any member of the coalition.

Formally,

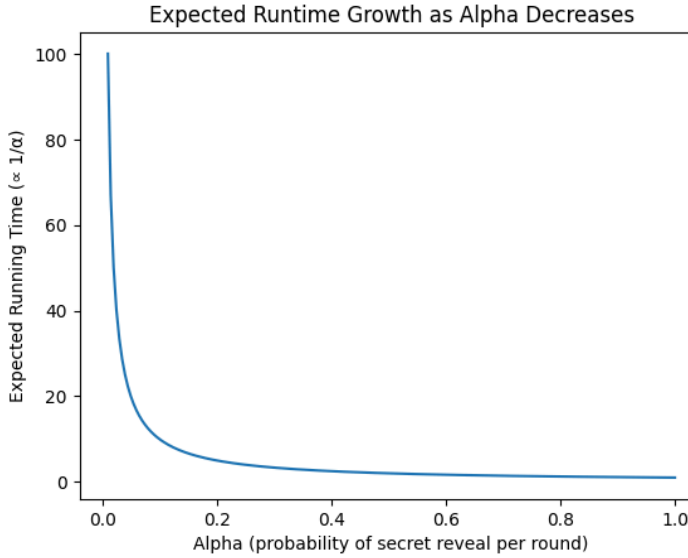
$$u_i(\sigma) \geq u_i(\tau_C, \sigma_{-C}) \quad (2)$$

for every coalition C , every deviation τ_C , and all $i \in C$. When $k = 1$, the definition reduces to a Nash equilibrium.

III. RATIONAL SECRET SHARING

Secret sharing distributes a secret among multiple players so that only authorized groups can reconstruct it. A common scheme is Shamir's secret sharing, where a secret is embedded in a polynomial and shares correspond to points on the polynomial.

In rational secret sharing, players may refuse to reveal their shares if doing so reduces their utility. Therefore, protocols must ensure that revealing shares is the rational choice.



A. Protocol Overview

We consider an m -out-of- n secret sharing scheme where at least m shares are required to reconstruct the secret. The protocol proceeds in rounds:

- 1) Players send their shares to a mediator.
- 2) The mediator generates a randomized polynomial related to the secret.
- 3) Players exchange verification messages.
- 4) If the polynomial reveals the secret, the protocol terminates.

Randomization ensures that players cannot predict exactly when the secret will be revealed, which discourages deviation.

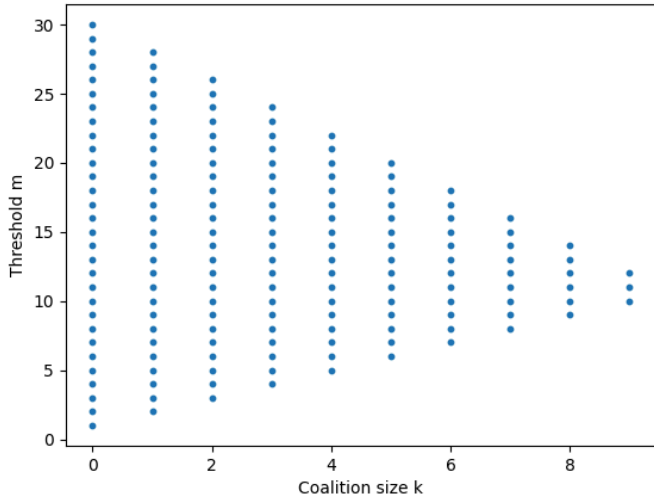
B. Resilience Guarantees

The protocol achieves a k -resilient equilibrium provided:

$$k < m \quad (3)$$

Under these conditions, no coalition of size k benefits from deviating. The expected running time of the protocol is $O(1/\alpha)$ rounds, where α controls the probability of revealing the secret in each round.

Feasible Region for k -resilient Secret Sharing ($k < m \leq n - 2k$, $n=3$)



IV. MULTIPARTY COMPUTATION

Secure multiparty computation (MPC) allows players to jointly compute a function over their private inputs without

revealing them.

Our results extend directly to MPC by combining circuit evaluation with rational secret sharing. Players first compute the function using a secure protocol and then distribute the result via the secret sharing mechanism.

If $3k < n$, we obtain a practical mechanism with expected running time of two rounds. More generally:

- If $2k < n$, resilient MPC is achievable with verification techniques.
- If $k < n$, cryptographic assumptions enable approximate resilience.

These bounds match classical distributed computing thresholds while incorporating rational behavior.

V. HANDLING UNEXPECTED UTILITIES

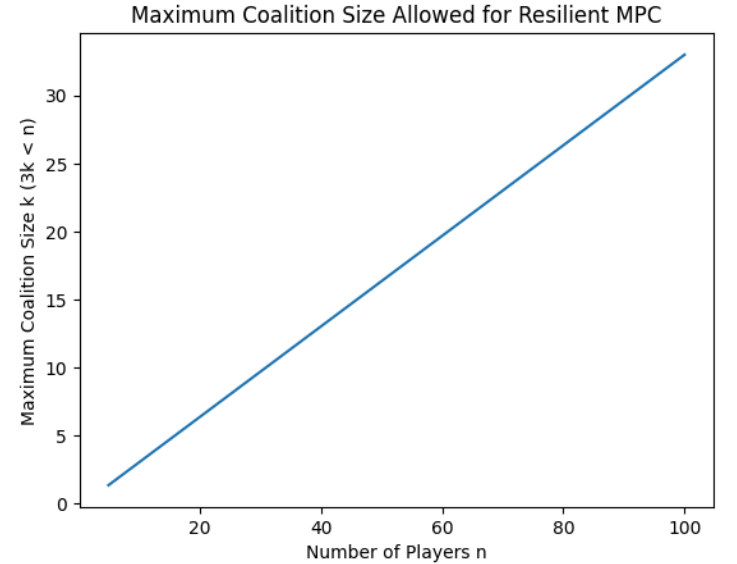
In large systems, some players may behave unpredictably or possess utilities different from those assumed by the protocol designer.

To address this issue, we introduce the notion of t -immunity. A strategy is t -immune if the utilities of rational players remain unaffected by the actions of up to t unpredictable players.

Combining this with coalition resilience yields (k, t) -robust equilibria. Such equilibria remain stable even when:

- up to k players form a rational coalition, and
- up to t players behave arbitrarily.

Protocols achieving (k, t) -robustness exist under conditions such as:



I.

SIMULATING MEDIATORS

Many equilibrium protocols rely on a trusted mediator that coordinates communication. However, mediators are often impractical in decentralized systems.

We show that mediator-based equilibria can be simulated using *cheap talk*, meaning direct communication between players.

The simulation works by replacing mediator actions with secure multiparty computation. Under suitable conditions, the resulting protocol preserves (k, t) -robust equilibrium properties.

This demonstrates that trusted mediators are not strictly necessary for achieving secure rational computation.

VII. CONCLUSION

This paper explores the integration of distributed computing and game theory in the context of rational agents. We introduced mechanisms for secret sharing and multiparty computation that achieve strong coalition resilience.

Our results show that:

- Rational secret sharing protocols can achieve strong equilibrium guarantees.
- Multiparty computation remains secure even with strategic participants.
- Protocols can tolerate unexpected utilities and arbitrary player behavior.
- Mediator-based equilibria can be simulated through direct communication.

$$3(t + k) < n \quad (4)$$

This result generalizes classical Byzantine fault tolerance models.

These findings contribute to the design of secure distributed systems where participants act strategically rather than maliciously.

Future work includes extending these mechanisms to asynchronous networks and large-scale distributed environments.

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